

Ethics & Governance Statement

Overview:

Given that this model is intended for the built environment, its deployment must conform to stringent ethical standards to ensure safety and maintain public trust.

- *Human-in-the-Loop (Safety):*

This model serves as a Decision Support Tool and is not intended to replace qualified structural engineers. All automated detections must be validated by a qualified professional prior to the authorization of any remedial measures or safety clearances. The risk associated with a "False Negative" (failing to identify a crack) bears considerable safety implications; therefore, the model is calibrated to prioritize high recall.

- *Data Privacy & Consent:*

The dataset was meticulously curated to exclude Personally Identifiable Information (PII). Any faces, license plates, or interior private spaces inadvertently captured during the imaging of facades have been blurred or removed to uphold the privacy of building occupants and pedestrians.

- *Bias and Fairness:*

I acknowledge the global diversity of construction materials and architectural styles. The model was developed using a comprehensive dataset to mitigate "material bias" (for example, recognizing cracks exclusively on smooth grey concrete while overlooking textured or painted facades). Continuous supervision is essential to ensure that the model performs effectively across various geographic regions and socioeconomic communities.

- *Transparency & Reproducibility:*

In accordance with the principles of Open Science, this project offers a comprehensive audit trail, which includes SHA256 checksums for verifying dataset integrity and a keyless execution environment. This approach allows third-party auditors to validate the model's performance without relying on concealed dependencies.